EEE 335 Laboratory 2
Assignment Description
Spring 2026

General Rules

- All reports need to be saved as PDF format.
- Use Report Submission Template to write out your lab report.
- In order to get full credit, you must add material indicated after the INCLUDE prompt under the tasks below.
- If you directly import cadence plots into your presentation, **they must have clearly visible axes, white background or CLEARLY labeled on black background, and solid easily viewable traces.**
- All schematics must be clearly annotated!
- All hand analysis may be scanned and pasted into the corresponding space(s) unless done directly on your electronic device (handwritten).
- Please ensure that your hand analysis follows the same naming scheme as your circuits/simulations.
- Please clearly mark all simulation axes (or title the simulation plots) (Grader does not know what V/net5 is!).

Lab Documents

This is the 'Assignment Description, it includes a brief overview of the lab as well as descriptions of all tasks students will have to take to complete this lab. For further information on how to complete these tasks reference the 'Lab2 Walkthrough' PowerPoint.

As the tasks are completed fill out the 'Report Submission Template' document. All graph titles that are required will already be populated; you need only add your results above them and your name to the upper right-hand corner. Once all blanks are filled, save the document as a pdf with the title being in the format "LastName_FirstName_Lab2.pdf".

Description

In this lab students will be creating and testing a CMOS inverter in Cadence. Students will be running DC and Transient analyses on the circuit and compare the results to their own hand calculations. During these steps, students will look at gate to source voltages, drain to source voltages, drain currents, propagation delays and noise margins. Students will then create a plot of the voltage transfer characteristic. Lastly students will run a Parametric analysis of length and width of the NMOS and PMOS and look at the changes to the Vin, Vout and VTC.

This lab consists of the following ten (10) tasks:

Task 1. Inverter DC Hand Analysis (1)

- Hand-calculate the values of V_{gs} (V_{sg}), V_{ds} (V_{sd}), & i_d for both NMOS and PMOS of your inverter when the input voltage is V_{IN} = 0V, 2.5V and 0.75V
- The following values may be used in your hand analysis:

$$k_p' = 95.75\mu A/V^2 \quad k_n' = 274.75\mu A/V^2$$
$$W_p = 1.35\mu m \quad W_n = 0.45\mu m \quad L_n = L_p = 0.3\mu m$$
$$V_{tp} \approx -0.54\,V \quad V_{tn} \approx 0.49\,V$$
$$Load\ Cap: C_L = 1pF \quad V_{dd} = 2.5\,V \quad V_{gnd} = 0\,V \quad V_{source_dc} = 0\,V$$

- INCLUDE: **Scan 1. DC Hand Calculations** of V_{gs} (V_{sg}), V_{ds} (V_{sd}), & i_d for NMOS and PMOS.

Task 2. Inverter DC simulation (2)

- Follow slide 7 of the Lab Walkthrough and make a DC simulation on the CMOS inverter you have built.
- INCLUDE: **Fig. 1. DC annotated schematic** (result screen-shot) as shown on slide 8 in EEE 335 Laboratory 2 Walkthrough for each input voltage.

Task 3. DC Conclusion and Comparison (2)

- On the Report Submission Template compare your results from the hand calculations and simulations.
- INCLUDE: **Table 1. Compare Hand Calcs & Simulation: Vgs, Vds, & id** for each input voltage. (Table 1a, 1b, and 1c)
- INCLUDE: A brief conclusion about your DC simulation.

Task 4. Inverter Transient Hand Analysis (4)

- Complete hand analysis and determine the rise time (t_r) & fall time (t_f), propagation delay Low->High (t_{PLH}), High->Low (t_{PHL}), average propagation delay (t_P), V_{OL}, V_{IL}, V_{IH}, V_{OH}, and noise margins N_{ML}, N_{MH} of the inverter.
- Both equations and results must be included. SHOW YOUR WORK.
- These are some parameters you can use in the calculation:

$$k_p' = 95.75\mu A/V^2 \quad k_n' = 274.75\mu A/V^2$$
$$W_p = 1.35\mu m \quad W_n = 0.45\mu m \quad L_n = L_p = 0.3\mu m$$
$$V_{tp} \approx -0.54\,V \quad V_{tn} \approx 0.49\,V$$
$$Load\ Cap: C_L = 1pF \quad V_{dd} = 2.5\,V \quad V_{gnd} = 0\,V \quad V_{source_dc} = 2.5\,V$$

- You can use R_p = 30kΩ for (W/L = 1) and R_n = 12.5kΩ for (W/L = 1)
- INCLUDE: **Scan 2. Transient Hand Calculations** of t_r, t_f, t_{PLH}, t_{PHL}, t_P, V_{OL}, V_{IL}, V_{IH}, V_{OH}, N_{ML}, and N_{MH}.

Task 5. Inverter Transient Simulation t_{PLH}, t_{PHL}, rise and fall time (4)

- Follow transient slides of the Lab Walkthrough and make a transient simulation on the CMOS inverter you have built.
- INCLUDE: Screenshot with the transient analysis plot, with rise/fall time, and propagation delays (t_{PLH}, t_{PHL}) clearly marked as shown on slides.
    - **INCLUDE two plots:**

      Fig. 2. Rise Time and Fall Time

      Fig 3. Propagation Delay Low to High and High to Low

Task 6. Inverter Transient Simulation Parametric Analysis (2)

- Follow slides 18, 19, 20 & 21 of the Lab Walkthrough and make a parametric analysis simulation on the CMOS inverter you have built.
- INCLUDE: **Fig. 4. Timing for a Sweep of Capacitance from 500fF to 1500fF**
- INCLUDE: Answer to the following question.
    - What value of capacitance has better rise and fall times and better propagation delay? Why is this true?

Task 7. Inverter DC Transfer Characteristics (2)

- Follow slides 23, 24, 25 & 26 of the Lab Walkthrough and make a DC transfer characteristics analysis simulation on the CMOS inverter you have built.
- INCLUDE: **Fig. 5. Voltage Transfer Characteristic** & mark NML & NMH.
    - Be sure to include the plots (3 windows) as shown on slide 26 of the walk through.

Task 8. Transient Conclusion and Comparison (2) from tasks 4, 5, and 7.

- INCLUDE:

  Table II. Compare Hand Calcs & Simulation: tPHL, tPLH, trise, tfall, NML & NMH
- INCLUDE: A brief conclusion about your Transient simulation.

Task 9. Inverter Transient Simulation Parametric Analysis-Length (2)

- Do the transient simulation and sweep the length of NMOS from 300nm to 700nm with 5 steps and keep L_pmos=300nm.
- Do the transient simulation and sweep the length of PMOS from 300nm to 700nm with 5 steps and keep L_nmos=300nm.
- INCLUDE: Screenshots for the parametric analysis on length of PMOS and NMOS.
- **Two plots**

  Fig. 6. Vin & Vout for a Sweep of NMOS Length from 300nm to 700nm

  Fig. 7. Vin & Vout for a Sweep of PMOS Length from 300nm to 700nm

Task 10. Inverter DC Transfer Characteristics Parametric Analysis-Length (4)

- Do the DC Transfer Characteristics simulation and sweep the length of NMOS from 300nm to 600nm with 2 steps and keep L_pmos=300nm, W_pmos=1.35um, W_nmos=450nm.
- Do the DC Transfer Characteristics simulation and sweep the length of PMOS from 300nm to 600nm with 2 steps and keep L_nmos=300nm, W_pmos=1.35um, W_nmos=450nm.
- Do the DC Transfer Characteristics simulation and sweep the width of PMOS from 1.35um to 2.7um with 2 steps and keep L_nmos=300nm, L_pmos=300nm, W_nmos=450nm.
- Do the DC Transfer Characteristics simulation and sweep the width of NMOS from 450nm to 900nm with 2 steps and keep L_nmos=300nm, L_pmos=300nm, W_pmos=1.35um.
- INCLUDE: Screenshots for DC Transfer Characteristics Parametric Analysis,
    - Four plots. Each plot is similar to plots on lower left side of slide 26.

      Fig. 8. VTC for a Sweep of NMOS Length from 300nm to 600nm

      Fig. 9. VTC for a Sweep of PMOS Length from 300nm to 600nm

      Fig. 10. VTC for a Sweep of PMOS Width from 1.35um to 2.7um

      Fig. 11. VTC for a Sweep of NMOS Width from 450nm to 900nm

- INCLUDE:
    - **Table III. Compare Varying Lengths and Widths: VIL, VIH, NML, & NMH**
- INCLUDE: A brief conclusion about your Parametric sweep.